

Graph Learning

Suggested Citation: Feng Xia, Ciyuan Peng, Jing Ren, Falih Gozi Febrinanto, Renqiang Luo, Vidya Saikrishna, Shuo Yu and Xiangjie Kong (2025), "Graph Learning", : Vol. XX, No. XX, pp 1–[181](#). DOI: XXX.

Feng Xia

RMIT University
f.xia@ieee.org

Ciyuan Peng

Federation University Australia
ciyuan.p@ieee.org

Jing Ren

RMIT University
jing.ren@ieee.org

Falih Gozi Febrinanto

Federation University Australia
f.febrinanto@federation.edu.au

Renqiang Luo

Jilin University
lrenqiang@outlook.com

Vidya Saikrishna

Federation University Australia
v.saikrishna@federation.edu.au

Shuo Yu

Dalian University of Technology
shuo.yu@ieee.org

Xiangjie Kong

Zhejiang University of Technology
xjkong@ieee.org

This article may be used only for the purpose of research, teaching, and/or private study. Commercial use or systematic downloading (by robots or other automatic processes) is prohibited without explicit Publisher approval.

now

the essence of knowledge

Boston — Delft

Contents

1	Introduction	3
2	Scalable Graph Learning	14
2.1	Graph Data Summarization	16
2.2	Computational Sampling Methods	26
2.3	Distributed Graph Learning	34
3	Temporal Graph Learning	37
3.1	Dynamic Graph Learning	38
3.2	Spatiotemporal Graph Learning	43
4	Multimodal Graph Learning	50
4.1	Graph-driven Multimodal Learning	51
4.2	Learning on Multimodal Graphs	57
5	Generative Graph Learning	64
5.1	Unconditional Generative Graph Learning	64
5.2	Conditional Generative Graph Learning	74
6	Explainable Graph Learning	78
6.1	Post-hoc Explanation for Graph Learning	79
6.2	Self-explanatory Graph Learning	84

7	Responsible Graph Learning	88
7.1	Privacy-Preserving Graph Learning	89
7.2	Robustness of Graph Learning	94
7.3	Fairness in Graph Learning	99
8	Emerging Topics	106
8.1	Graph Foundation Models	107
8.2	Graph Reinforcement Learning	112
8.3	Federated Graph Learning	114
8.4	Learning on Knowledge Graphs	118
8.5	Knowledge-infused Graph Learning	122
8.6	Quantum Graph Learning	126
9	Conclusion	128
	Acknowledgements	130
	References	131